Remote Sensing of Environment 244 (2020) 111793

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

An integrated methodology using open soil spectral libraries and Earth Observation data for soil organic carbon estimations in support of soil-related SDGs

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ARTICLE INFO

Keywords:
Soil organic carbon
Hyperspectral imagery
Soil spectral library
Memory-based learning
Sustainable development goals

ABSTRACT

There is a growing realization amongst policy-makers that reliable and accurate soil monitoring information is required at scales ranging from regional to global to support ecosystem functions and services in a sustainable manner under the amplifying climate change enabling countries in target setting of the Sustainable Development Goals (SDGs). In this line, the need of access to and integration of existing regional in situ Earth Observation (EO) data and different sources such as contemporary and forthcoming satellite imagery is highlighted. The current study puts major emphasis on leveraging existing open soil spectral libraries and EO systems and bridging them with memory-based learning algorithms that create more cost-efficient and targeted large scale mapping of soil properties. Relying mostly on contemporary capacities and open resources it can be readily applied to countries with differing capacities and levels of development. To test our methodology, the GEOCRADLE SSL developed in the Balkans, Middle East, and North Africa region and a hyperspectral airborne image were utilized to provide Soil Organic Carbon (SOC) maps of cropland fields over an agricultural region near the city of Netanya, Israel. Furthermore, simulated data of forthcoming space-borne satellite (EnMAP) and current super-spectral mission (Sentinel 2) were explored. The SOC content of the collected in situ soil samples was predicted using a novel local regression approach that combines spatial proximity and spectral similarities. These predictions were subsequently used to develop models using the airborne and simulated satellite spectra, achieving a fair prediction accuracy of $R^2 > 0.8$ and RPIQ > 2.

1. Introduction

The introduction of the 2030 Agenda for Sustainable Development offers a unique opportunity to ensure food, water and energy security, protect biodiversity, and mitigate climate change towards clear guidelines and targets. At the same time, land degradation and declining of organic matter have direct consequences for the deterioration in environmental quality and undermine the wellbeing across several countries (Bouma and Montanarella, 2016; Cowie et al., 2018). In this context, the multifunctionality of soils is increasingly recognized as a key enabler in the management of other natural resources, and researchers focused on the provision of accurate maps (Viscarra Rossel et al., 2019). Although the soil is not a Sustainable Development Goal (SDG) in itself, the sustainable management and monitoring of soils towards achieving SDGs have been recognized as important in previous studies (Tóth et al., 2018), which identified ten targets with direct synergies with soil across several goals.

Considering the world demand for monitoring and reporting of soil related SDGs, there is an estimation that millions of soil samples should be analyzed worldwide on an annual basis (Demattê et al., 2019). Recent advancements in remote sensing, and particularly in hyper- and multi-spectral technologies, have brought it to the forefront as an effective tool for substantial land mapping and monitoring. Thus, the enforcement and monitoring of soil related SDG indicators requires the establishment of integrated management systems supported by high level geospatial products derived from these forthcoming technologies (Keesstra et al., 2018). In this context, there is a growing realization amongst policy makers and researchers that Earth Observation (EO) can

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https://doi.org/10.1016/j.rse.2020.111793
Received 16 November 2018; Received in revised form 2 January 2020; Accepted 23 March 2020
Available online 29 April 2020
0034-4257/

serve as a core component towards addressing the increasing global demand to collect, analyse and process soil data into formats that are relevant and actionable for the implementation of the SDGs, through a more simple, rapid and accurate way (Anderson et al., 2017). For instance, researchers increasingly rely on airborne imaging and ground data collection campaigns to establish associations between the reflectance spectra in the Visible to Near-InfraRed (VNIR) and Short-Wave InfraRed (SWIR) region and the targeted soil properties (Ben-Dor et al., 2002; Hbirkou et al., 2012; Steinberg et al., 2016), with the most prevalent soil property examined being the Soil Organic Carbon (SOC) concentration (Ben-Dor et al., 2009; Guo et al., 2019). However, these approaches based on imaging spectroscopy usually account only for relatively small regions since they entail time-consuming and expensive data processing activities on the field scale (extensive sampling and laboratory analysis), as well as sophisticated and expensive infrastructure and have thus low replicability to other regions (Chabrillat et al., 2019). Therefore, the systematic exploitation of airborne hyperspectral data across wide regions remains largely a difficult task, due to either lack of appropriate monitoring systems, or disparate level of laboratory infrastructure, or lack of funding for operational campaigns across the different countries, coupled with the ineffective exploitation of available complementarities in expertise. Consequently, airborne campaigns cannot adequately assist the monitoring and reporting of SDGs at the country level.

The digital soil mapping, based on space borne EO data, is currently undergoing a significant shift. This is driven by the advent of the big EO data era, mainly spearheaded by Sentinel-2 free and open super spectral imagery data. A critical review about recent advances in remote sensing techniques for soil monitoring was provided by Angelopoulou et al. (2019). In that regard, recent studies (Castaldi et al., 2019a; Taghadosi et al., 2019; Vaudour et al., 2019), have shown the way forward through the exploitation of Sentinel-2 data, together with advanced regression analytics for estimating and mapping the spatial variability of soil properties, such as SOC, over a single considered date. Previously, Gholizadeh et al. (2018) extracted spectral signatures from Sentinel-2 bands, from four independent study sites involving different types of agricultural areas and used a support vector regression algorithm at pixel level to produce spatial distribution maps of soil properties. Each pixel was enriched by some additional features coming from combinations performed amongst the extracted band values. In addition to the Sentinel archive, multispectral imagery data from the Landsat program has also been utilized into a promising framework to map soil texture based on a powerful data mining procedure to retrieve spectral reflectance signatures (Demattê et al., 2018). The new hyperspectral sensors, such as the recently launched PRecursore IperSpettrale della Missione Applicativa (PRISMA, Galeazzi et al., 2008) or the forthcoming Environmental Mapping and Analysis Program (EnMAP, Guanter et al., 2015) as well as other on track (e.g. Surface Biology and Geology VNIR Mission, Lee et al., 2015), will provide unprecedented data streams for the retrieval and hence monitoring of soil variables, across the VNIR–SWIR spectral range, with increased representativeness and reliability.

Along with the growing availability of EO data, the open-access soil spectral libraries (SSLs) have enabled a data-driven approach to effectively describe soil, both qualitatively and quantitatively, by finding robust statistical associations between spectral signatures and soil properties (Ballabio et al., 2016; Nocita et al., 2015). As part of achieving that goal, there has been a large effort and considerable investment in the soil science community to compile extensive data sets of soil properties and provide stakeholders with access to soil information. The Land Use and Coverage Area Frame Survey (LUCAS) SSL was developed as a continuously evolving database by the European Union and is considered the largest harmonized open-access dataset of topsoil properties at global scale (Orgiazzi et al., 2018; Panagos et al., 2012). Viscarra Rossel et al. (2016) have proposed the development of an international voluntary database of vis-NIR spectra (~20,000 soil spectra) linked to information on the soils' composition. The need for the application of standard measurement protocols and internal soil standard methods amongst the collaborating actors, to optimally allow the processing of the collected set of heterogeneous data due to diverse sources and spectrometers, towards the enhancement of predictive performance has been highlighted in previous studies (Kopačková and Ben-Dor, 2016; Romero et al., 2018). Consistent with these efforts, a study by Tsakiridis et al., 2018 was able to develop a standardized spectroscopic database of existing legacy soil data archives by applying internal soil standard protocols in spectral measurements (Ben Dor et al., 2015). This latest activity was in the direction of enhancing the still fragmented and uncoordinated in situ component of the Global Earth Observation System of Systems (GEOSS).

The study of VNIR–SWIR point and imaging spectroscopy has grown enormously in the recent years, assisted by novel developments in the domain of machine learning (Tsakiridis et al., 2017, 2019a, 2019b). Recognizing the necessity to exploit open contemporary SSLs, efforts in the past years have focused on incorporating this prior information to local sites, to enhance the prediction accuracy at the local site. Consistent with this, Ramirez-Lopez et al. (2013) proposed the use of high performance local regression models, which utilize the spectral similarity to identify the most similar set of samples from a large SSL to each local sample. This set is then used to develop the multivariate regression model. The so-called spiking approach, where a small subset of samples from the local site is utilized to augment the larger SSL was also examined in previous studies (Guerrero et al., 2014). Moreover, Lobsey et al. (2017) proposed an automatic way to find the most representative subset of the larger SSL to assist the predictions in the smaller site, while in a recent work Castaldi et al. (2019b) provided a promising framework to define the appropriate sampling pattern and number of samples based on the feature space of ENMAP data.

However, such models developed from laboratory spectra cannot be readily applied to spectra from remote sensors, considering that no reliable transfer function exists to date. The necessity in designing effective integration strategies of data from satellite and airborne with ground-based measurements is therefore an important problem, which was further recently highlighted by the European Union's Copernicus programme. In this context, Castaldi et al. (2018) proposed a bottom-up approach to map soil properties using hyperspectral data with the LUCAS SSL without the need for site-specific sampling collection. In addition, Liu et al. (2018) investigated the potential of fine-tuning a previously trained convolution neural network using a small subset of samples from the local site to obtain a more precise model. Although these methods pave the way for an integrated approach to estimate soil variables over large areas, as will be required once the hyperspectral satellites will be available to all, these studies did not evaluate the optimal number of local samples that should be selected as a subset for spiking to enhance the predictions (Guerrero et al., 2014). Moreover, only a few studies report experiments concerning the use of forthcoming hyperspectral imagery data (Steinberg et al., 2016). And finally, no attempt is usually made to use or combine different spectral preprocessing methods which are known to enhance the predictive performance (Stenberg et al., 2010).

In this line, the present study aims to: (i) present a new monitoring scheme, termed spiked bottom-up approach, and its crucial requirements for integrating information from existing open SSLs and spectral imagery data, towards delivering quantitative and spatially explicit soil indicators; (ii) report on the development of a novel Local Gaussian Regression (LGR) method, part of the spiked bottom-up approach, utilizing memory-based learning that combines different spectral pretreatments and develops site-specific models using local subsets from a calibration dataset; and (iii) evaluate the proposed scheme at field sites with contrasting pedological zones, whereby the assets of current and forthcoming spectroscopic imagers will be explored, based on simulated data from airborne imagery.

The novel element of the proposed scheme is that it enables a significant cost reduction for monitoring and reporting soil related indicators without sacrificing the accuracy, by performing laboratory analyses only to a small subset of site-specific samples, since it leverages data mining techniques for handling and integrating ground spectroscopic and remotely sensed imagery data with existing soil database infrastructures. It thus paves the way for a systematic baseline determination of several key soil properties to support countries in target setting of the Sustainable Development Agenda.

2. Materials

2.1. Study area — local site

The study took place at the Ha-Ogen plain located northeast of the city of Netanya, at 32°21′23.15″ N and 34°57′28.24″ E, Israel (Fig. 1), and the data acquisition was carried out in 2013 (Ogen et al., 2017). The Ha-Ogen plain is considered an agricultural area, where land use is dominated by orchards of citrus fruit and avocado trees. The climate of the area is Cas according to the Köppen–Geiger climate-classification system (Kottek et al., 2006), with a mean annual precipitation of 570 mm and a dry summer. The Ha-Ogen plain can be divided in two agro-pedological zones which are characterized by the presence of two soils types, respectively. The first one is mainly characterized by brown-red sandy soil (Hamra in local definition system) which correlates well with Rhodoxeralfs accompanied with the Aquic Rhodoxeralf, (Nazaz soil subgroup in local definition system) – and the other one is characterized by Vertisol – Haploxererts (Soil Survey Staff, 2014).

2.2. Soil spectral libraries

The main scientific challenge of this study was to evaluate the utility of VNIR–SWIR spectroscopy (both point and imagery) to produce accurate predictions of soil properties supported by the integration of the information from hyper- and super spectral imagery with the one provided by already existing soil spectral data archives. The first step entailed the use of a small representative set of site-specific soil data and of a recently established VNIR–SWIR SSL, hereafter referred to as local SSL and GEOCRADLE SSL (GSSL), respectively. For more details (e.g. sampling strategies and soil types) the reader is referred to Ogen et al. (2017) and Tziolas et al. (2019), respectively. Hereafter, only their main characteristics are presented for a better understanding.

The local SSL consisting of 45 samples was assembled by pooling together data from topsoil samplings (0–30 cm) collected from eight agricultural fields under an intensive soil sampling procedure (2 to 14 samples per field). The field data campaign was performed simultaneously with the airborne data acquisition, following general guidelines, standards, and protocols to ensure consistent data collection.

The GSSL contains 1582 soil samples and their corresponding SOC measurements, across nine countries in the North Africa, Middle East, and Balkan region. The data in the library are diverse, comprising soils of 18 soil classes of the world. Approximately 58% of the soil samples belong to the top soil layer (0–30 cm), 20% at depths within 30–60 cm, and 20% from 60 to 100 cm, while the rest 2% originate from samples collected at depths > 100 cm. The soil samples were collected with geographical coordinates in the World Geodetic System (WGS84).

The soil samples (both local SSL and GSSL) were air-dried, crushed, sieved and the fine earth (< 2.0 mm) was transported to the laboratory for chemical and spectroscopic analysis. The Walkley-Black method was applied for the measurement of SOC (Heanes, 1984).

Both SSLs contain VNIR–SWIR standardized spectral signatures (350–2500 nm). For each measurement, ten spectra were averaged to minimize noise and to maximize the Signal-to-Noise Ratio (SNR). Spectra were recorded with a sampling resolution of 1 nm so that each spectrum comprised reflectance values at 2151 wavelengths. The local SSL and GSSL were created using two different spectrometers (350–2500 nm), an Analytical Spectral Devices–ASD Fieldspec Pro (PANanalytical BV, Boulder, CO, USA formerly Analytical Spectral Devices) and a PSR+ spectrometer (Spectral Evolution Inc., Lawrence, Massachusetts,USA), respectively. It should be noted that we implemented standard measurement protocols and internal soil standard methods (Ben Dor et al., 2015) for both local and regional SSL development to ensure (i) the elimination of potential spectral measurement variations due to different acquisition techniques, instruments and protocols, and (ii) the fusion into similar international initiatives, such as the global SSL (Viscarra Rossel et al., 2016). The GSSL is in compliance with GEOSS data principles and Open Database License standards. The library is publicly available through the GEOCRADLE's project regional datahub.[1]

2.3. Airborne hyperspectral imagery acquisition and processing

Remote sensing data acquisition with airborne hyperspectral sensor (AisaFenix, Specim, Spectral Imaging Ltd., Finland) was carried out over the fields whereas a single continuous hyperspectral cube (380–2500 nm), was acquired on September 17th 2013 (Fig. 1), at an aircraft altitude of 700 m above ground level, resulting in a spatial resolution of 1 m. The AisaFenix sensor acquires data in push-broom mode covering the VNIR-SWIR spectral range (380–2500 nm) with 620 spectral channels, having a FWHM of 3.5–5.5 nm, FOV of 32.3° and high SNR (600–1000). The flight lines covered a total area of about 0.75 km^2.

To fine tune calibrate the data radiometrically, the supervised vicarious calibration procedure as proposed by Brook and Ben Dor (2011) was implemented and then we obtained the surface reflectance for the subsequent steps. In order to identify the bare soil pixels, the airborne imagery was masked out for other surfaces such as manmade structures, water bodies and vegetation by applying the spectral angle mapper algorithm, as described by Ogen et al. (2017).

Subsequently, to reduce the effect of noise in the AisaFenix data, the SNR of the full 620 channels was examined. We removed bands with a SNR lower than 100 and NIR bands affected by strong atmospheric water absorption, retaining 344 bands (88, 138 and 118 bands in the VIS, NIR and SWIR regions, respectively) for the rest of the analysis.

2.4. Simulated imaging data

The airborne hyperspectral reflectance image was used to produce synthetic spectra for the Sentinel-2 and EnMAP satellite missions (Sentinel-2 was not operational at the time of the airborne campaign). To ensure that the produced spectra are consistent with the spectral and radiometric characteristics of the different analyzed satellite imagers, we followed the procedure detailed in Castaldi et al. (2016). More specifically, the airborne hyperspectral spectra were resampled based on convolution procedures using the spectral response function of the Sentinel's superspectral imager, while the one of EnMAP was approximated using Gaussian functions. Both resampling procedures considering the spectral configurations are described in the relevant work of Gomez et al. (2018). The spatial resampling has not taken into account in this study.

3. Methods

3.1. Integrated monitoring scheme for SOC estimation – the spiked bottom-up approach

In order to estimate the soil properties by reducing the analytical cost and thus improving the financial viability of soil spectroscopy, the spiked bottom-up approach is proposed in this paper building upon the works of Castaldi et al. (2018) and data spiking (Brown, 2007). It

[1] http://datahub.geocradle.eu/dataset/regional-soil-spectral-library.

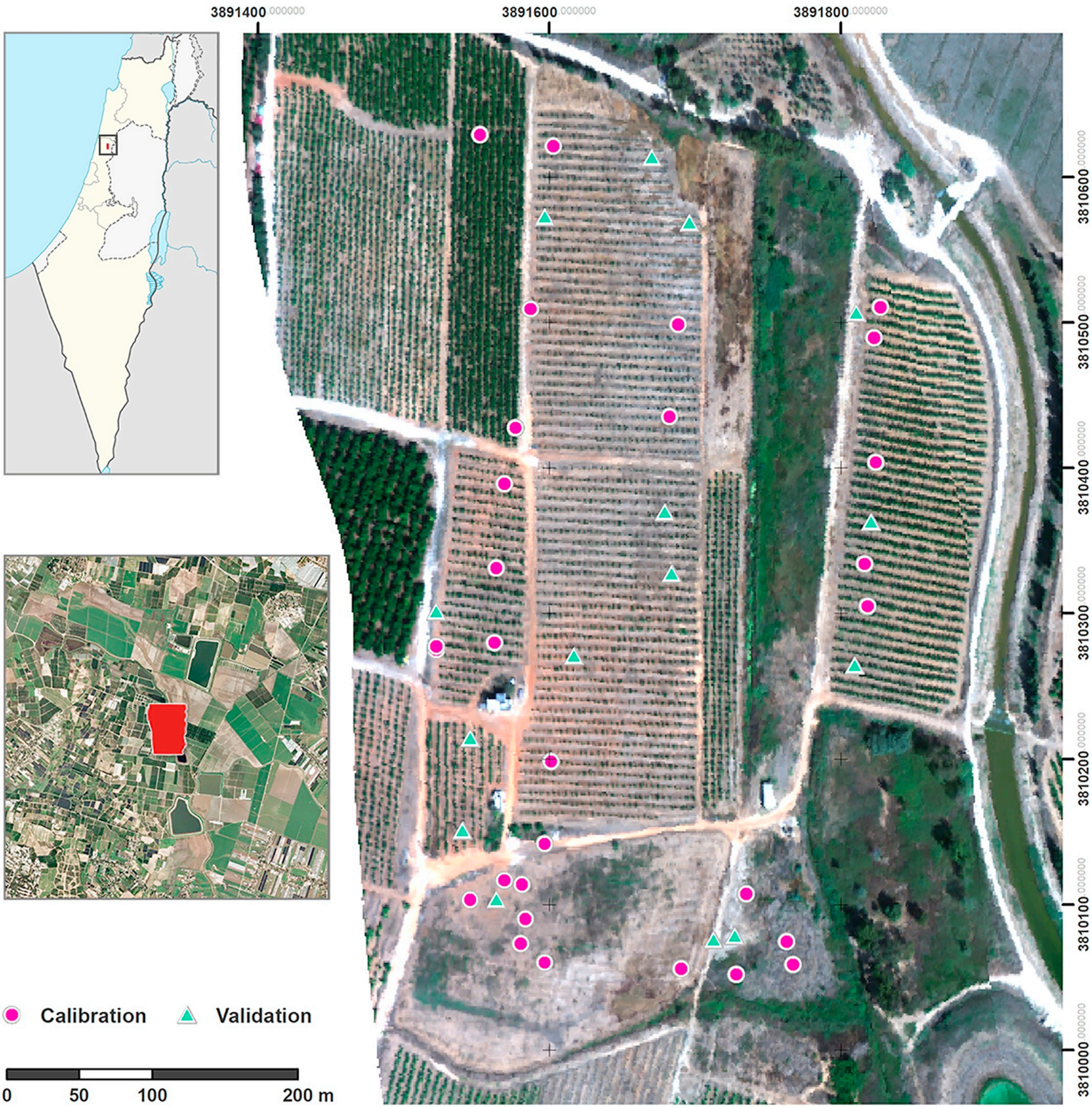

Fig. 1. Map of Israel the location of the study area, and the RGB representation of the hyperspectral image collected by the AisaFenix sensor over the Ha-Ogen plain from 17th September 2013, depicting the location of the sampling points.

derives local spectroscopic calibrations of soil properties, based on the novel LGR algorithm, by combining existing information in a spectral library with a representative set of site-specific samples.

3.1.1. Overview of the spiked bottom-up approach

The proposed scheme (Fig. 2) involves two main steps at the local site: (i) prediction of soil properties using laboratory point spectroscopy at the site-specific samples; and (ii) prediction of soil properties using imaging spectroscopy across the whole site.

The first step entails the estimation of the target SOC in the sampling points while trying to minimize the cost of analytical measurements. To this end, the soil samples from the pilot area were split into two disjoint sets, with the first smaller set containing representative in terms of spectral information samples, identified using the Kennard-Stone algorithm (Kennard and Stone, 1969). This set then underwent laboratory analysis for the estimation of SOC and was used to augment the regional SSL (spiking approach), thereby assisting LGR in the more accurate predictions of the second set.

In the second step of the spiked bottom-up approach, the aforementioned estimated values in addition to the ones measured analytically (i.e. from all samples of the local site) were used to calibrate a model for the estimation of the SOC concentration of the pixels in the spectroscopic image (as derived by any EO sensor) corresponding to the sampling points. Accordingly, a LGR model was applied for the bare soil

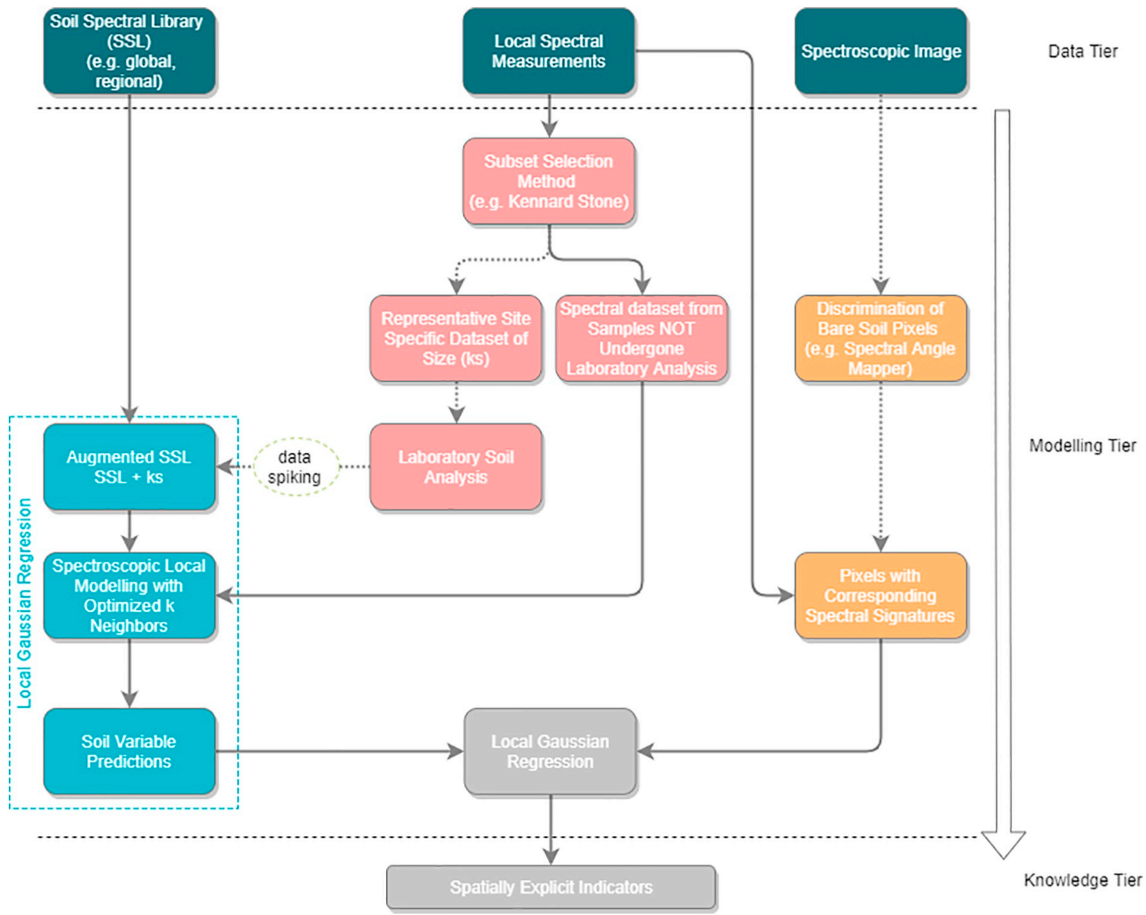

Fig. 2. Schematic diagram concerning the derivation of spatially explicit soil carbon indicator based on the proposed spiked bottom-up approach. The dotted lines indicate that the spiking step is not obligatory to be implemented.

pixels in each area and was subsequently used to generate SOC maps.

3.1.2. The proposed modified Local Gaussian Regression (LGR) process

Given an existing reference SSL comprised of N samples with known outputs $D_{\mathrm{cal}} = \{\mathbf{Xr}, \mathbf{yr}\} = \{\mathbf{xr}_{\mathrm{i}}, yr_i\}_{i=1}{}^{i=N}$, the goal is from a set of M unknown samples $D_{\mathrm{test}} = \{\mathbf{Xu}, \mathbf{yu}\}$ to predict the unknown outputs $\mathbf{yu}$ using $\mathbf{Xu}$. Instead of building a global model and applying it for all samples in D_{test}, LGR constructs a local model for each unknown sample using only its closest neighbors, following the memory-based learning approach. Assume that the reference predictors $\mathbf{Xr}$ and unknown predictors $\mathbf{Xu} = \{\mathbf{xu}_{\mathrm{i}}\}_{i=1}{}^{i=M}$ are defined in an initial spectral space $\mathscr{X}_1$ (e.g. the initial reflectance spectra). These spectra are transformed using a set of predefined spectral treatments with the goal to maximize the spectral information so that a set of p spectral spaces $\{\mathscr{X}_i\}_{i=1}^{i=p}$ is defined. For a given $\mathbf{xu}_i$ and to infer its predicted output yu_i, LGR works by selecting a set of k closest neighbors from D_{cal} by utilizing a spatio-spectral distance, i.e. by taking into account both the geographical proximity and the spectral similarity in one of the spectral spaces $\mathscr{X}_1^o$. It then performs a Gaussian regression using a cosine kernel calculated in a second spectral space $\mathscr{X}_2^o$ using only the set of closest neighbors. All the parameters that need to be optimized for LGR are identified through an internal cross-validation procedure in D_{cal}. In the paragraphs that follow, more details governing the LGR algorithm are given.

3.1.2.1. Pre-processing techniques. A common practice in soil spectroscopy is to test different pre-processing techniques which may help to enhance the absorption bands or perform scatter correction. A detailed list of such techniques may be found in Rinnan et al. (2009). The following set of commonly applied pre-processing techniques was examined herein: (i) reflectance spectra (Ref), (ii) reflectance spectra plus standard normal variate (Ref-SNV), (iii) reflectance spectra plus de-trending (Ref-DT), (iv) the pseudo absorbance transformation (Abs), (v) smoothing of the absorbance spectra using a zero-order Savitzky-Golay filter and standard normal variate (Abs-SG0-SNV), (vi) first-derivative absorbance spectra using a first order Savitzky-Golay filter (Abs-SG1), (vii) first-derivative absorbance and standard normal variate (Abs-SG1-SNV), (viii) second-derivative absorbance spectra using a second order Savitzky-Golay filter (Abs-SG2), (ix) the continuum removal transformation (CR).

3.1.2.2. Distance calculation. The distance between $\mathbf{xu}_{\mathrm{i}}$ and $\mathbf{xr}_{\mathrm{j}}$ is calculated by taking the average between the spatial (Euclidean) distance, and the spectral distance in $\mathscr{X}_1^o$; each of these two distances are normalized to [0, 1] before the average is calculated. As far as the spectral distance is concerned, it is calculated using the Mahalanobis distance in the optimized Principal Components (PC) space (Ramirez-Lopez et al., 2013). The optimum number of PCs is selected through leave-one-out cross-validation in D_{cal}, where for each number of PCs the prediction accuracy is estimated using the 1-Nearest-Neighbor (1-NN) approach (Cover and Hart, 1967).

3.1.2.3. Prediction – Kernel computation. LGR uses as predictors both the spectral information in $\mathscr{X}_2^o$ as well as the local distance matrix.

To predict the output, LGR employs Gaussian Process regression (Williams and Rasmussen, 1996) with a cosine kernel. For a given set of predictors $\mathbf{X}$ the cosine kernel function is the L2-normalized dot product defined for two row vectors $\mathbf{x}_1$, $\mathbf{x}_2$ as:

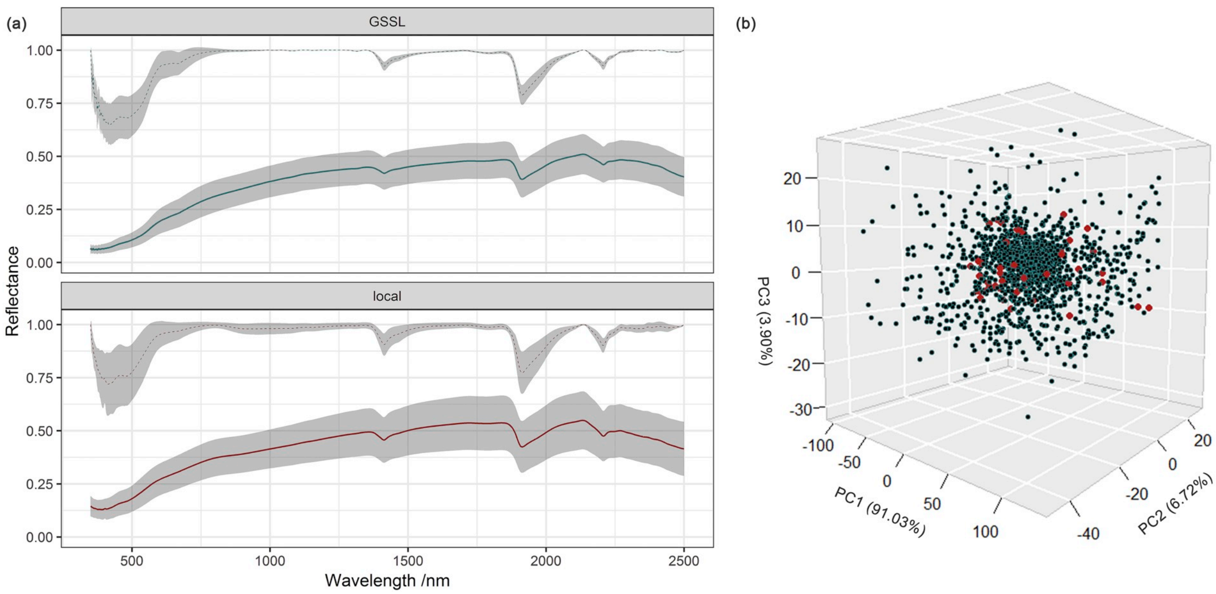

Fig. 3. The average reflectance spectrum of the GEOCRADLE and local SSLs, upper curves are the continuum removed spectra and b) projection of the site-specific spectra from Israeli study site (local SSL, red dots), on to the first three principal component scores of the GSSL (black-cyan dots). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$K(\mathbf{x}_1, \mathbf{x}_2) = \frac{\mathbf{x}_1\mathbf{x}_2^T}{||\mathbf{x}_1||\,||\mathbf{x}_2||} \tag{1}$$

3.1.2.4. Optimizing the number of neighbors. To find the optimal k, denoted as k_o, an internal cross-validation procedure is followed that resembles the leave-one-out cross-validation. For each $\mathbf{xu}_i$ its closest neighbor in D_{cal} is determined. Then, with this sample removed from D_{cal}, LGR predicts its (known) output using its k closest neighbors in D_{cal}. The optimal k_o is the one which minimizes the prediction error. An overview of this procedure is presented in Algorithm 1.

```
Require: D_cal = {Xr, yr}, D_test = {Xu, yu}, {X_i}_{i=1}^{i=p}
Ensure: Optimal LGR configuration
for each pair (X_1, X_2) ∈ {X_i}_{i=1}^{i=p} × {X_i}_{i=1}^{i=p} do
    Find oPC using leave-one-out and 1-NN in X_1 of D_cal
    Calculate the pairwise spatio-spectral distance matrix D using Euclidean
  spatial distance, and oPC-Mahalanobis distance in X_1 between all pairs in
  Xr ∪ Xu.
    for k = k_start, ..., k_max do                              ▷ optimize k
        for xu ∈ Xu do                               ▷ for each unknown sample
            Find its nearest neighbor in D_cal denoted as xru using D
            Find the k nearest neighbors of xru from D_cal using D
            Extract a local distance matrix of size k × k from D containing
  the pairwise distance of the neighbors
            Predict ŷru using a Gaussian Process Regression with a cosine
  kernel using as predictors the local distance matrix and the spectra in X_2.
        end for
        Calculate RMSE of the predictions
    end for
end for
Identify the optimal configuration: spectral pair (X_1^o, X_2^o), oPC, and k_o
  according to the RMSE
```

3.2. Validation and verification of the results

To validate the proposed methodology the initial 45 samples were split into calibration (30) and validation (15) using random sampling within each field, as depicted in Fig. 1. In order to ensure that we have a similar SOC range the samples were sampled from the different quartiles. The validation set was independent and used only to assess the performance of the models. The calibration set represents the collected samples which will be used to calibrate the models, i.e. the local spectral measurements as indicated in the data tier of Fig. 2.

The accuracy of the predictive models was finally evaluated by examining the Root Mean Square Error (RMSE, Eq. (2)), the coefficient of determination (R^2; Eq. (3)), and the Ratio of Performance to Interquartile Range (RPIQ; Eq. (4)). The RPIQ is commonly used as a better indication for non-normal variables (Bellon-Maurel et al., 2010). The equations used were as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{i=N}(y_i - \hat{y}_i)^2}{N}} \tag{2}$$

Table 1
The distribution of soil organic C (%) in the GSSL and at the Israeli test site. Q25, Q75, Sd, and Skew indicate the first and third quartile, the standard deviation and skewness, respectively.

Dataset	n	Min	Q25	Median	Mean	Q75	Max	Sd	Skew
GSSL									
All	1582	0.00	0.78	1.12	1.23	1.51	6.38	0.80	2.10
Local									
Calibration	30	0.15	0.91	1.37	1.76	2.07	6.07	1.43	1.60
Validation	15	0.27	0.93	1.39	1.95	2.22	6.42	1.63	1.64

$$R^2 = 1 - \frac{\sum_{i=1}^{i=N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{i=N} (y_i - \bar{y})^2} \tag{3}$$

$$RPIQ = \frac{IQ}{RMSE} \tag{4}$$

where y_i is the observed value and $\hat{y}_i$ is the predicted value of sample i, N the number of observations (Eq. (2)), $\bar{y}$ is the mean of the observed values (Eq. (3)), and IQ is the interquartile range (IQ = Q3 − Q1) of the observed values (Eq. (4)).

Finally, as far as the hyperparameter optimization of LGR is concerned, the number of optimal PCs was searched in [1, 20], while k in [10, 15, …, 100] and the optimal values were selected according to the ones minimizing the RMSE of the validation set.

4. Results

4.1. Overview of the SSLs

As a result of the standardization process a harmonized dataset was generated. Fig. 3 illustrates the differences amongst the spectra from the site-specific soil samples (local) and of the large GSSL. Fig. 3a depicts the recorded reflectance spectra overlaid with the continuum removed ones (Rinnan et al., 2009), while Fig. 3b projects the local SSL spectra on to the first three principal components of the GSSL. Taking into account that different instruments and operators were involved for recording the spectra, the spectral signatures of the local SSL are nevertheless similar with the respective GSSL and quite indistinguishable, thereby enabling the subsequent multivariate analysis to potentially achieve high predictive performance. This also highlights the importance of the standardization procedure that we followed.

The chemical laboratory analysis concerning the SOC contents is presented for both local SSL and the GSSL in Table 1. It can be observed that the distribution of SOC in the local SSL falls largely within the respective GSSL one, following a positively skewed distribution with a similar range. This attests to the suitability of the GSSL to be used to infer the SOC contents of the local SSL.

Table 2
Results in the independent set for different values of k_s using the laboratory spectra, while k_o denotes the number of neighbors from the joint local and existing SSL.

ks	k_o	$\mathscr{X}_1^o$	$\mathscr{X}_2^o$	RMSE	R^2	RPIQ
0	60	Ref-DT	SG1	0.8951	0.6748	1.4412
5	45	SG0-SNV	SG1	0.8542	0.7039	1.5102
10	45	SG0-SNV	SG1-SNV	0.7629	0.7638	1.6909
15	35	Ref-DT	SG1	0.5353	0.8837	2.4097
20	30	SG0-SNV	SG1-SNV	0.5165	0.8917	2.4976
25	35	Ref-DT	SG1-SNV	0.5782	0.8643	2.2311
30	30	SG0-SNV	SG1-SNV	0.4561	0.9156	2.8283

4.2. SOC estimation using LGR and existing regional SSL

The first step in the spiked bottom-up approach entails the development of the augmented dataset, by incorporating selected samples from the local calibration site to the existing regional GSSL. To this end, the initial 30 samples of the calibration set needed to be split into two disjoint sets, with the first set of size *ks* being analyzed in the chemical laboratory and merged to form the augmented SSL. Ideally this number must be small in order to minimize the additional sampling and associated cost. To demonstrate the effect of *ks*, i.e. of the number of samples selected using the Kennard-Stone algorithm that need to be analyzed, we examined different values from the range {0, 5, …, 30}, with 0 representing the case where spiking is not employed, while 30 the case where all samples are used. For each value, a new LGR model was optimized according to Algorithm 1 and was subsequently used to estimate SOC content in the independent set. The results are presented in Table 2, with k_o denoting the number of neighbors used. It can be then observed that the model attains a good performance (i.e. RPIQ > 2) with ks = 15, where the average number of neighbors that LGR uses from the combined augmented SSL is 35 and thus the model efficiently uses both samples from the local dataset and from the existing SSL. The larger values of *ks* do not necessarily enhance the model's performance, with the best value (ks = 30) attaining a slight increase of performance for double the number of samples that need to be analyzed. It should further be noted that in all occasions the optimal pair of spectral sources $\mathscr{X}_1^o$ and $\mathscr{X}_2^o$ uses two different pre-treatments, thus highlighting the need for using the complementary information contained in different spectral sources. Overall, the results indicate that a small value of *ks* is enough to enhance the predictions of the LGR model. A scatter plot of the best model as derived with ks = 15 between the real and predicted values in the independent set is illustrated in Fig. 4.

4.3. SOC estimation using data from remote sensors

Using the optimal values obtained at the previous step, at this stage all soil samples have an associated SOC value, with 15 samples measured directly in the chemical laboratory and 15 predicted by the LGR model using the laboratory spectra. These values can be subsequently used as ground truth to develop the models from the EO spectral data using a second LGR model, to assign a SOC value to each soil pixel in the image. Due to the incorporation of the spatial similarity in the neighborhood selection of LGR and given that the described approach is pixel-based, it is expected that the so-termed salt-and-pepper noise might be reduced. The calibration of the LGR model occurs as described in Algorithm 1, where instead of using the laboratory spectra, now the predictors are (i) the recorded spectra from the hyperspectral sensor, (ii) the simulated EnMAP spectra, and (iii) the simulated Sentinel-2 spectra. The only difference is that in the case of the Sentinel-2 spectra and due to its limited number of broad spectral bands we only considered the spectral pre-treatments that were valid, i.e. not the ones employing a Savitzky-Golay filter.

The simulated EnMAP and Sentinel-2 soil spectra, alongside the original AisaFenix hyperspectral signatures, are depicted in Fig. 5. This figure demonstrates that the simulated EnMAP and the recorded AisaFenix soil spectral signatures are quite similar, with the first having fewer channels and appearing to be comprised of smoother spectra, while there is a significant amount of information not retained by Sentinel-2. Regardless, to ascertain their effectiveness, their prediction accuracies must be assessed.

Table 3 shows the predictive performance of the best model as derived by LGR per each of the EO sensors. It can be noticed that, as expected, the best performance is obtained by the hyperspectral airborne image which exploited the wide spectral information content of the AisaFenix sensor. Despite that, the simulated EnMAP results follow close behind with the simulated Sentinel-2 data attaining relatively

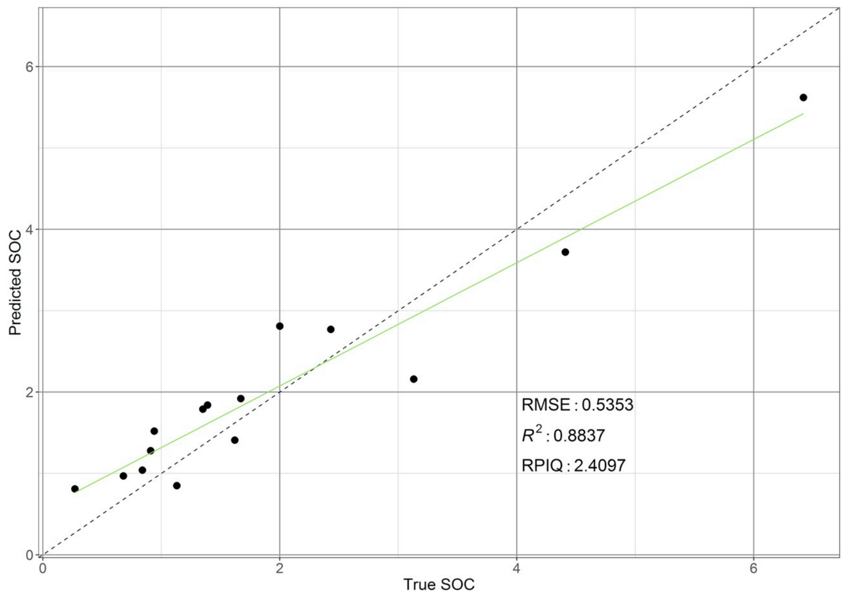

Fig. 4. Scatter plot between observed and predicted SOC for the independent set of soil samples from the local site, using the laboratory spectra and ks = 15.

satisfactory results. For both EnMAP and the AisaFenix datasets the LGR model effectively combines the same two spectral sources, namely SG0-SNV and SG1-SNV. The results aforementioned can be explained by the fact that LGR strives to maximize the information contained in the calibration models by utilizing different spectral sources for an optimal neighbor selection and regression analysis.

Fig. 6 illustrates a comparison of SOC maps as derived by the spiked bottom-up approach. The spatially explicit indicators representing SOC created by the airborne AisaFenix and simulated EnMAP and Sentinel-2

Table 3
Results in the independent set for the different remote sensing imagery data based on LGR models.

EO image	k_o	$\mathscr{X}_1^o$	$\mathscr{X}_2^o$	RMSE	R^2	RPIQ
AisaFenix	30	SG0-SNV	SG1-SNV	0.6087	0.8496	2.1194
EnMAP	30	SG0-SNV	SG1-SNV	0.6170	0.8455	2.0907
Sentinel-2	30	Ref-SNV	Ref-SNV	0.6285	0.8397	2.0525

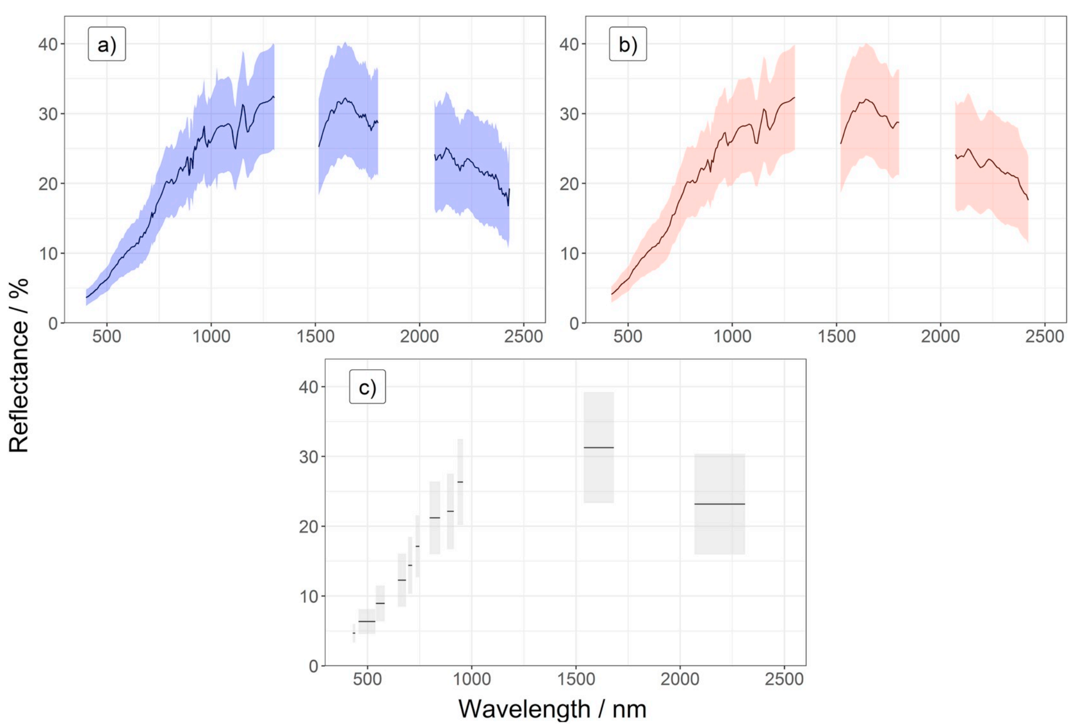

Fig. 5. a) Mean ± standard deviation of the recorded soil spectra of the AisaFenix sensor and of the b) simulated EnMAP and c) Sentinel-2 soil spectra. The bands with noise in the water absorption ranges were removed.

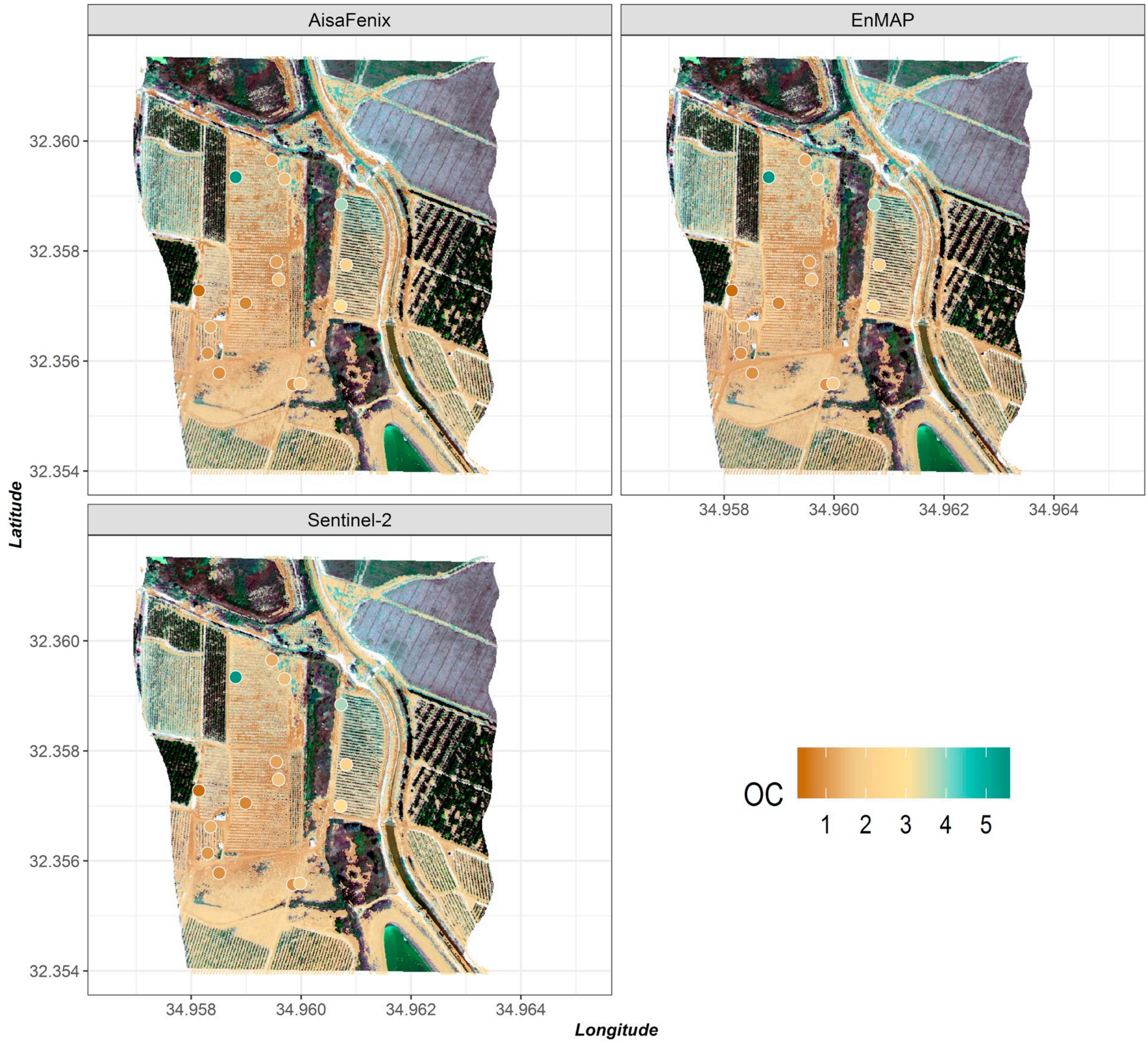

Fig. 6. SOC maps as generated by the LGR algorithm applied to the different imagery data overlaid with the original RGB bands of AisaFenix hyperspectral data cube. The spatial resolutions of the simulated Sentinel-2 and EnMAP were not resampled to their actual pixel size.

data showed generally similar trends in spatial distribution, within the study region. It can be noticed that there exists a clear north-eastern gradient in SOC content, with lower values in the areas with alluvial sediments, which in overall is in consistence with the distribution of the real values.

4.4. Comparisons with other approaches

A comparison between the spiked bottom-up methodology and other approaches has been performed. In particular we implemented two different approaches (i) the traditional approach and (ii) a bottom-up approach without spiking of site-specific samples.

In the traditional approach, all collected calibration samples (30) were sent to the chemical laboratory to be analyzed. Afterwards, a global model was built using the hyperspectral imagery spectra as predictors, and the analyzed values as target property. To this end, as far as the predictors are concerned, we tested all the aforementioned spectral pre-treatments, while the following commonly employed machine learning algorithms were used: (i) PLS, (ii) SVM, and (iii) Cubist (analytical description of the algorithms can be found in the study of Viscarra Rossel and Behrens, 2010).

Subsequently, we evaluated the performance of the bottom-up approach, as described in the general frame of Fig. 2 (dotted lines) without any site specific samples used in the calibration set (i.e. using only the GSSL as the calibration set), and as indicated in Table 2, where m = 0.

Table 4 summarizes the descriptive statistics of the performance of the different approaches in order to produce SOC maps. The best results for the traditional approach were attained using the SVM algorithm and the absorbance spectra (see Supplementary material). However, Fig. 7 shows that the SVM algorithm assigns to the majority of the pixels a low SOC value, close to the median, indicating that it has not successfully modelled the association between spectral values and target property.

5. Discussion

5.1. Evaluation of the spiked bottom up approach

In this study we present the spiked bottom-up approach which exploits contemporary open SSLs and hyperspectral EO data, to produce comparative quantitative assessments and corresponding mapping over large geographical areas, thereby assisting countries with different

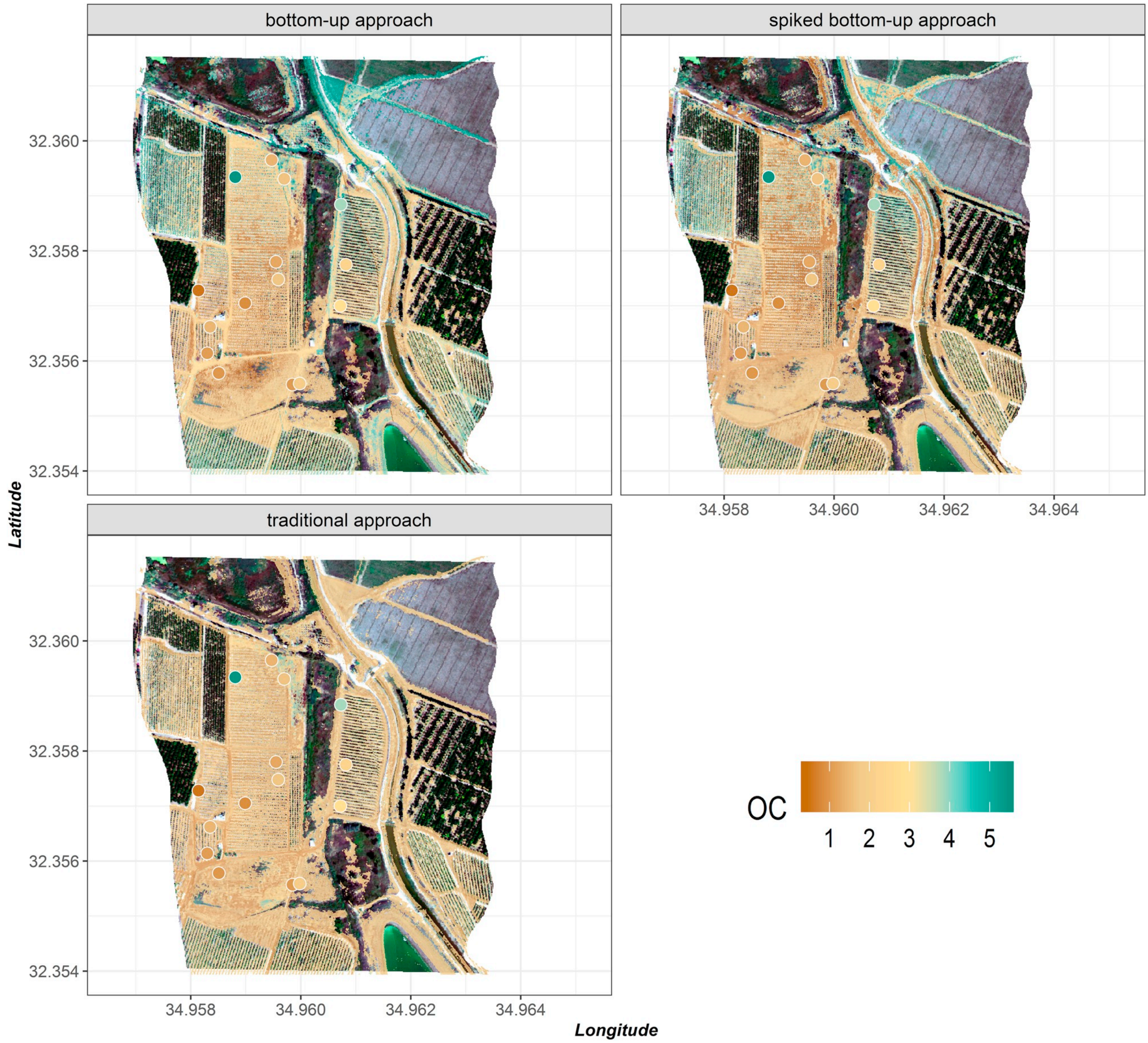

Fig. 7. Comparison maps of SOC as generated by the (i) bottom-up, (ii) spiked bottom-up (based on the LGR algorithm) and (iii) traditional (based on the SVM algorithm and using all available calibration samples) approaches from the AisaFenix image, with the points indicating ground soil samples used for validation purposes.

levels of development to report progress towards soil related SDGs. As highlighted in previous studies (Nocita et al., 2015; Viscarra Rossel et al., 2016), significant challenges hamper the stability of the remote-sensing data and, consequently, the alignment between standardized laboratory measurements and remote spectra, under natural conditions. Thus, the proposed approach is a two-stage process involving the following steps: (i) predicting soil properties from laboratory spectral signatures, and (ii) generating spatially explicit indicators of the targeted properties using EO imagery data. The modelling step in both cases is implemented by the novel LGR algorithm.

Overall, the main advantage of the spiked bottom-up as compared to the traditional approach is that the chemical analyses in the calibration dataset are either completely replaced by spectral measurements ($ks = 0$) or depend on a subset of the site-specific samples ($ks > 0$). The value of ks may be selected from the end-user depending on the accuracy needed and the financial constraints. As illustrated in the results of Table 2, LGR models with $ks < 15$ led to lower accuracy than models with $ks \geq 15$, probably due to fewer suitable neighbouring spectra for robust subset generation. These results are encouraging and suggest that LGR can reduce the number of calibration soil samples that

Table 4
Descriptive accuracy metrics of the performance of the different approaches to generate SOC maps from the AisaFenix on the independent test set; m denotes the number of samples from the local dataset used in the calibration procedure of each approach.

Approach name	Calibration set	Model	m	RMSE (%)	R^2	RPIQ
Bottom-up	GSSL	LGR	0	0.9234	0.6539	1.3970
Spiked bottom-up	GSSL + k_s samples from the local SSL	LGR	15	0.6087	0.8496	2.1194
Traditional	All local samples	SVM	30	0.8317	0.7193	1.5510

need to be analyzed by conventional laboratory analytical methods, without sacrificing the accuracy of prediction.

To our best knowledge, there are limited studies combing the use of existing SSLs and remote spectroscopic data (Castaldi et al., 2018), as well as only few studies have specifically compared the capability of current (Sentinel-2) and forthcoming (EnMAP) satellite imagers, within this framework. In the current study, the use of AisaFenix and LGR to estimate SOC results in fair prediction accuracy ($R^2 > 0.8$, RPIQ > 2) as shown in Table 4. The SOC prediction performance obtained here are slightly higher than those previously obtained by Vaudour et al. (2016, RPD 1.3–1.6) as well as by Castaldi et al. (2018, RPD = 1.7) both utilizing airborne hyperspectral imagery data, acquired over slightly larger areas. The results obtained by LGR are also better than model performances obtained with the traditional approach (Table 4). This is attributable to the generation of representative local subsets by the LGR algorithm, which confirms the importance of combined spectral sources and geographical proximity in the selection of closest neighbors (Tziolas et al., 2019). As indicated in Fig. 6, higher values were obtained in the north-west part, which is in good agreement with the distribution of the validation samples. It can be inferred that the proposed methodology produces more accurate predictions and maps while also simultaneously minimizing the analytical cost, based on the comparison of Fig. 7. What is more, the incorporation of the spiking samples is straightforward, while the extraction of the relevant subset of samples from the regional SSL occurs automatically for each unknown sample and is not subject to exhaustive search as in other approaches. Equally important is the development of more elaborate sampling strategies for the collection of the appropriate local calibration samples. In a recent work, Castaldi et al. (2019b) provided a promising framework to define the appropriate sampling pattern and number of samples based on the feature space of ENMAP data. These outputs indicated the need to adopt appropriate sampling strategies to collect a calibration dataset representative of the region of interest and simultaneously including, as much as possible the variability of the soil property. In this respect, future extensions may focus on developing methodologies for the automatic determination of *ks* i.e. of the number of samples used for spiking, using e.g. active learning techniques.

Previous studies (Castaldi et al., 2019a; Gholizadeh et al., 2018) using only Sentinel-2 data demonstrated that its spectral characteristics are adequate to predict the organic carbon content of the top soil in croplands (RPD = 1.00–2.60). Vaudour et al. (2019) also obtained mediocre results for SOC prediction using Sentinel-2 data in two agricultural plains (RPD = 1.00–1.51). These satisfactory performances of SOC prediction by Sentinel-2 sensor, also, are in compliance with the current study (Table 3) and may be explained by the fact that the SOC absorption feature is represented by three Sentinel-2 spectral bands at B4 and B5 followed by B11 and B12. Similarly to the studies aforementioned, the comparison between multispectral and hyperspectral data has not remarkable differences in terms of SOC prediction accuracy. Based on these findings the loss of spectral resolution, due to the resampling procedure, is probably balanced by the improvement of signal to noise ratio. In this context, the coarsening of spectral resolution did not cause a large decrease in the model performance for SOC estimation.

The standardization procedure was another important aspect of this study. Considering that the efficiency of the bottom-up approach is strongly affected by the existing large SSLs, it becomes clear why the standardization step is crucial. In contrast to similar studies where the site-specific samples and the existing SSLs were measured with the same instruments (Liu et al., 2018), the soil samples used in our study were analyzed with different spectral instruments but under specific protocols. Therefore, the proposed methodology is not tied to a specific instrument. Our results suggest that spectral standardization protocols could be a significant step in order to facilitate the access and integration of any new soil spectral measurement into existing SSLs irrespective of the instrument and laboratory used.

Admittedly, the present study has a few shortcomings. Firstly, it was evaluated only in one test site and limited number of site-specific samples were used; therefore, more extensive testing, including real space borne data, is necessary to ascertain the robustness of the methodology. Moreover, the estimation accuracy obtained by AisaFenix could not be completely comparable with EnMAP and Sentinel-2 simulated data, because the AisaFenix data has a finer spatial resolution than the other sensors investigated in this work (30 m and 10 m, respectively). In addition, the SNR values of the simulated sensors might be different from the AisaFenix SNR used in this study and hence might affect the results accordingly. Spatial resolution influences the prediction accuracy of soil variables (Gomez et al., 2015), however the impact of the various spatial resolution was not investigated in this study and needs further research.

5.2. The case for large scale soil mapping using affordable techniques in response to SDGs

Soil has a fundamental role to play in efforts to ensure food security and land degradation neutrality, since it is straightforward connected with productivity terms (SDGs 2.3, 2.4, 15.3). Furthermore, soils fulfil a large number of functions and ecosystem services (Adhikari and Hartemink, 2016; Keesstra et al., 2016) that explicitly bind them with other goals including SDGs 3.9, 6.4, 6.5 (soil-water holding capacity and conductivity); SDG 11.3 (reference to land resources); and SDG 13.2 (climate regulation through SOC sequestration).

In the light of freely available data of current multispectral sensors (e.g. Landsat, Sentinel-2), a virtual constellation of these data sources can be evaluated to include more important spectral features related to other soil properties. In the next few years, a wide range of satellite EO systems will become available strengthening soil monitoring to reach the desired spatio-temporal representativeness and reliability. In particular, the forthcoming hyperspectral sensors in orbit, such as the Environmental Mapping and Analysis Program (EnMAP, Guanter et al., 2015), Surface Biology and Geology VNIR Mission and SHALOM (Feingersh and Dor, 2015) in addition to PRISMA recently placed in orbit, will soon provide unprecedented data streams for the retrieval and hence monitoring of soil variables, across the VNIR–SWIR spectral range. Concurrently, a series of novel and potentially low-cost observational platforms, unmanned aerial vehicles (Aldana-Jague et al., 2016), and cubesats are rapidly maturing and becoming available to carry out the next generation of hand-sized hyperspectral imagers.

In recognition of the significant disparities and the different maturity levels in terms of monitoring capacities across the countries, an easily deployable and cost-effective EO approach to provide baseline determination of key properties to support countries in the monitoring and reporting of soil related SDGs needs to be developed. The Group on Earth Observation recognized these requirements from its initiation, with the insistency on improved collaboration between scientific groups but also by seeking to break down both capacity building and data sharing barriers. Concurrently, to the existing GEOSS, we should consider the establishment and operation of an in-situ hub to facilitate the access to data, models, protocols and extraction of information that enables the development of a holistic approach to large scale soil mapping, capable of achieving a global baseline monitoring system.

5.3. Future work and suggested next steps

Further work should focus on the development of more elaborate sampling strategies for the collection of the appropriate local calibration samples as well as automatically ascertaining the number of local samples to be used for spiking. In addition, applying the proposed approach in more areas and larger scales for predicting other soil variables, such as soil texture elements, pH and CEC, in order to ascertain its robustness, is deemed necessary.

Another interesting effort may be to develop a strategy to cope with measurements directly based on in-situ spectra, taking into account the effects of water and other environmental factors utilizing novel handheld sensors. More work is also needed on how to best adapt the proposed approach in non-bare fields for deriving soil property maps by overcoming the spectral perturbation posed by partial vegetation coverage. In addition, further research is required in the applications of platforms enabling large scaling processing of geospatial data, such as the Google Earth Engine (Gorelick et al., 2017) and the Committee on Earth Observation Satellites Data Cube (Giuliani et al., 2017). These platforms can facilitate in the construction of single synthetic images which would represent soil using multi-temporal image information as indicated by Diek et al. (2016) and Demattê et al. (2018), overcoming the issues of partial cloud coverage. The current approaches, including this one, emphasize in spatial prediction of properties that are relatively static over the observational time period. This raises issues relating to the combined influence of surface roughness and moisture of bare soils on spectral reflectance, especially for the quantitative assessment of soil parameters. It may be interesting, in future studies, to estimate the impact of these hindrances on the prediction performance and leverage the growing trend in applications of deep learning in two-dimensional predictions based on spectral and temporal features.

6. Conclusion

In this study, we present the spiked bottom-up monitoring framework to derive SOC estimations in support of soil-related SDGs. In particular, we focus on the development of a memory-based learning algorithm able to derive local calibration models of any soil property leveraging information from existing SSLs and a small set of site-specific samples. As demonstrated, spiking the existing SSLs with a set of representative site-specific samples can significantly increase the accuracy of the predicted properties. Additionally, the proposed method produced noticeable gains in accuracy and cost-efficiency in all examples compared to other state of the art methods, with a satisfactory level of accuracy ($R^2 > 0.8$, RPIQ > 2). Furthermore, we demonstrate the versatility of our technique by using it regardless of the availability of site-specific samples (ks = 0) taking into account the recognized gaps and capacities in less developed countries. We noted that the number of the site-specific samples (ks) in the spiked bottom up approach is very crucial for the final accuracy (RPIQ = 1.44–2.82) and should therefore be specified in future studies based on more elaborate techniques (e.g. active learning). We show that the proposed approach can integrate the laboratory spectroscopic predictions and hyper- and super spectral EO measurements towards delivering quantitative and spatially explicit indicators. The spiked bottom up approach as applied to AisaFenix data achieved the overall best results (RPIQ = 2.1194) compared to simple bottom up (RPIQ = 1.3970) or the traditional approach (RPIQ = 1.5510) and can enhance the SOC prediction accuracy compared to other state of the art machine learning models. This is attributable to the generation of representative local subsets by the LGR algorithm, which confirms the importance of combined spectral sources and geographical proximity in the selection of closest neighbors from a SSL. In addition, the prediction accuracies obtained by hyperspectral AisaFenix and simulated EnMAP data (with the same SNR), are very close to that obtained by Sentinel-2 data despite the loss of spectral resolutions. Our vision is that this methodology could be used as a basis for quickly creating models based on the information contained in various open SSLs and, thereby, initiate a step forward for large scale soil mapping based on the current and forthcoming satellite imagery data in supporting and tracking the progress of the SDGs.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2020.111793.

Acknowledgements

This paper is based on the work that has been conducted in the framework of the GEOCRADLE and E-SHAPE projects, which have received funding from the European Union's Horizon 2020 research and innovation programme under Grant Agreement No 690133 and No 820852, respectively.

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